

```

10. NUM_DIGITS = 3 # (!) Try setting this to 1 or 10.
11. MAX_GUESSES = 10 # (!) Try setting this to 1 or 100.
12.
13.
14. def main():
15.     print('Bagels, a deductive logic game.
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17.
18. I am thinking of a {}-digit number with no repeated digits.
19. Try to guess what it is. Here are some clues:
20. When I say:      That means:
21.   Pico           One digit is correct but in the wrong position.
22.   Fermi          One digit is correct and in the right position.
23.   Bagels         No digit is correct.
24.
25. For example, if the secret number was 248 and your guess was 843, the
26. clues would be Fermi Pico.''.format(NUM_DIGITS))
27.
28.     while True: # Main game loop.
29.         # This stores the secret number the player needs to guess:
30.         secretNum = getSecretNum()
31.         print('I have thought up a number.')
32.         print(' You have {} guesses to get it.'.format(MAX_GUESSES))
33.
34.         numGuesses = 1
35.         while numGuesses <= MAX_GUESSES:
36.             guess = ''
37.             # Keep looping until they enter a valid guess:
38.             while len(guess) != NUM_DIGITS or not guess.isdecimal():
39.                 print('Guess #{}: '.format(numGuesses))
40.                 guess = input('> ')
41.
42.             clues = getClues(guess, secretNum)
43.             print(clues)
44.             numGuesses += 1
45.
46.             if guess == secretNum:
47.                 break # They're correct, so break out of this loop.
48.             if numGuesses > MAX_GUESSES:
49.                 print('You ran out of guesses.')
50.                 print('The answer was {}'.format(secretNum))
51.
52.             # Ask player if they want to play again.
53.             print('Do you want to play again? (yes or no)')
54.             if not input('> ').lower().startswith('y'):
55.                 break
56.         print('Thanks for playing!')
57.
58.
59. def getSecretNum():
60.     """Returns a string made up of NUM_DIGITS unique random digits."""
61.     numbers = list('0123456789') # Create a list of digits 0 to 9.
62.     random.shuffle(numbers) # Shuffle them into random order.
63.
64.     # Get the first NUM_DIGITS digits in the list for the secret number:

```